HACK-A-ROO 2026 PITCH-TO-BUILD TRACK | RESEARCH PROTOTYPE • SIMULATED PATIENTS • NOT FOR CLINICAL USE

Hypertension Alert System for Pregnancy

Protocol Tracking, Unit and Bedside Views, and Traceable EMR Notes for Maternal Care

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1. Problem Statement

60 min
National standard, from the first severe reading

69%
Of persistent events treated within it on the pitch team's unit (their figure)

Where the time goes: the pitch team found delays between the first severe reading, confirming that it persists, notifying the right physician, and treatment. Much of that relies on attention and memory, while staff manage their other work.

- **Severe range:** systolic ≥ 160 mmHg or diastolic ≥ 110 mmHg.
- **Persistent:** severe again 15–60 minutes after the first reading, or no reading below the severe range by 15 minutes.
- **60-minute target:** counted from the *first* severe reading, not from confirmation.

Project Objectives

- **Rules that do the remembering:** time every step in tested rules, with no AI between a severe reading and an alert.
- **Unit and bedside views:** show the whole floor at once, and each bed's countdowns.
- **Traceable EMR note:** draft it for the nurse to review, with every time and number checked against the record.
- **Evaluation:** show what a given speed-up would be worth, with the speed-up stated as an assumption.

2. Datasets & Synthetic Patients

Results use **synthetic patients**. No protected health information is used.

80
Patients, 10 in each of 8 scenarios

50
With a severe reading

30
With no episode

- **Severe courses:** persistent, non-persistent, recurrent, spurious, and delayed or missing repeat.
- **No episode:** non-hypertensive, mild hypertension, borderline.

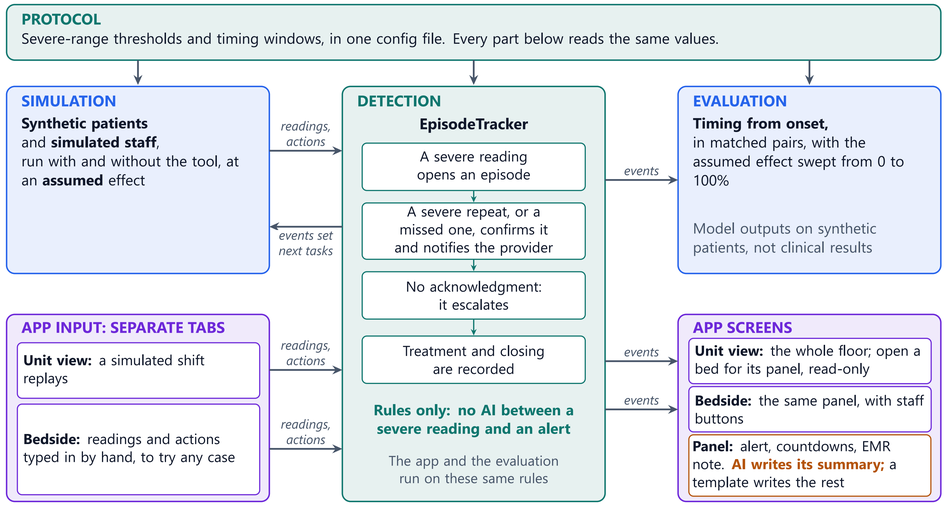
Prototype also includes a loader and benchmark patient scenario for UCI's Maternal Health Risk dataset (1,014 records; Ahmed et al., 2020; CC BY 4.0).

3. The Episode Rules

`EpisodeTracker` follows SMFM's persistent-episode definition (Combs et al., 2022), with timings in one config file:

- **Opened:** a severe reading starts the 60-minute clock.
- **Repeat due:** 15 minutes after the first severe reading.
- **Confirmed:** severe again in the window, or a missed repeat, so nobody has to remember it's due. The provider is notified.
- **Escalated:** whichever comes first, 10 minutes without acknowledgment or 10 minutes before the deadline. Both are our assumptions.
- **Reopened:** severe again within the hour is the same episode.

4. System Architecture



5. The App: Unit and Bedside Views

SIMULATED TIME **08:32** *Unit view, as the app shows it: an illustrative floor and mix of synthetic patients*

LDR 1 *No episode* **117/77** Routine	LDR 2 *Persistent by default: no recheck in time (SMFM C, D, F, G)* **135/85** Resolved Closed 08:30	LDR 3 *No episode* **159/105** Routine	LDR 4 *Persistent (SMFM E1): treated in time* **162/104** Confirmed Treatment target 09:00
corridor · Nurses' station			
LDR 5 *No episode* **147/95** Routine	LDR 6 *Persistent (SMFM E1): not treated in time* **168/106** Escalated Treatment target 09:00	LDR 7 *Not persistent (SMFM A)* **118/81** Resolved Closed 08:15	LDR 8 *Severe again within the hour: same episode (SMFM B)* **176/116** Repeat pending Repeat due 08:45

Unit view: one card per bed. LDR 4 and LDR 6 both confirmed at 08:15; LDR 4's provider responded and LDR 6's did not, so it escalated at 08:26. Open a bed for its panel. **Bedside:** the same panel with staff buttons, for readings typed in by hand.

6. Traceable EMR Note

An **LLM writes the note's summary** (Claude, or a local model through Ollama), and a template writes the timeline from the episode's events. Drafting runs outside the alerting loop.

The Traceability Rule: every time, number and patient identifier in the note must trace back to a system `Event` or a protocol value. A summary that fails is dropped: the next model tries, or the note goes on without one.

- **No diagnosis, no treatment advice:** the model is instructed not to give either.
- **Nurse review:** it's a draft for the nurse. The prototype connects to no real EMR.

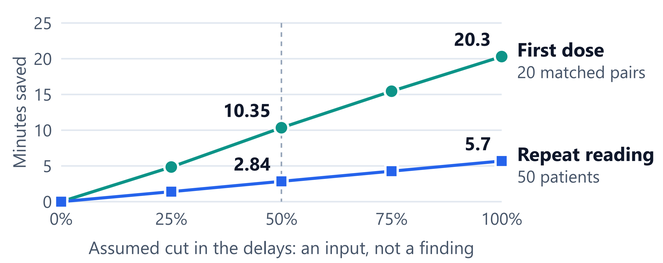
7. Paired Simulation: Model Outputs

The same 80 synthetic patients, simulated without and with the tool, **at an assumed 50% cut** in the time to notice a reading, repeat it, and reach a provider:

Measure	Without	With	Change
First dose, min from first severe reading (20 matched pairs)	66.20	55.85	**10.35 sooner**
Repeat reading, mean min late (50 patients)	5.70	2.86	**2.84 sooner**
Timely, of persistent episodes (denominators differ)	17 of 42	23 of 41	
of which by a dose	5	12	
of which by documented resolution	12	11	

What these can't show: the effect is an input, not a finding, so they say what a speed-up would be worth, not that the alert delivers one. Shortening the search for a provider is itself an assumption, and escalation gets no benefit in the model. The model also cancels a dose once the patient's underlying BP improves, even before a reading shows it. Not clinical results, not from any particular clinic, and not comparable with the 69%.

8. Minutes Saved Depend on the Assumed Effect



9. Key Contributions & Future Work

- **Rules can do the remembering:** every step is timed, aimed where the pitch team found the delays.
- **One set of rules:** the app and the evaluation run on the same tracker, covered by 616 passing tests.
- **Separation:** rule-based alerts, with AI limited to the note's summary.
- **Future work:** a far better simulation, timings observed on a real unit, modelling how a provider is found, HL7/FHIR monitor feeds, EHR integration, and evaluation in simulation labs.

See It Run

Watch the video
devans2718.github.io/hasp-video

Try the app
devans2718.github.io/hasp-app

Read the code
github.com/devans2718/hypertension-alert-pregnancy

Combs CA et al. Society for Maternal-Fetal Medicine Special Statement: A quality metric for evaluating timely treatment of severe hypertension. *Am J Obstet Gynecol* 2022;226(2):B2-B9. • Icons: Tabler Icons (MIT)